Functions & Transformations

ANSWER KEY

Section 1: Function Notation, Domain & Range

1 If $f(x) = 3x^2 - 2x + 5$, what is the value of $f(-2)$?

a. 13

b. 17

c. 21 ✓

d. 25

Substitute $x = -2$: $f(-2) = 3(-2)^2 - 2(-2) + 5 = 3(4) + 4 + 5 = 12 + 4 + 5 = 21$.

2 What is the domain of the real-valued function $g(x) = \sqrt{2x - 8}$?

a. $x \geq 4$ ✓

b. $x > 4$

c. $x \leq 4$

d. $x \geq 0$

The expression under the radical must be non-negative: $2x - 8 \geq 0 \Rightarrow 2x \geq 8 \Rightarrow x \geq 4$.

3 The function h is defined by $h(x) = \frac{7}{x-3} + 4$. For what value of x is $h(x)$ undefined?

a. -3

b. 0

c. 3 ✓

d. 4

A rational function is undefined when its denominator equals zero: $x - 3 = 0 \Rightarrow x = 3$.

Section 2: Function Transformations

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4 The graph of $y = f(x)$ is shifted 4 units to the right and 3 units down. Which equation represents the transformed function $g(x)$?

a. $g(x) = f(x - 4) - 3$ ✓

b. $g(x) = f(x + 4) - 3$

c. $g(x) = f(x - 4) + 3$

d. $g(x) = f(x + 4) + 3$

Horizontal shift right by h replaces x with $x - h$. Vertical shift down by k subtracts k. Thus $g(x) = f(x - 4) - 3$.

5 If the graph of $f(x) = x^2$ is reflected across the x-axis and vertically stretched by a factor of 5, what is the resulting equation?

a. $y = -5x^2$ ✓

b. $y = 5(-x)^2$

c. $y = -\frac{1}{5}x^2$

d. $y = -(5x)^2$

Reflection across the x-axis multiplies by -1; vertical stretch by 5 multiplies by 5. The equation is $y = -5x^2$.

6 The function $f(x)$ has a local minimum at $(2, -5)$. What is the coordinates of the corresponding local minimum on the graph of $y = f(x + 3) + 4$?

a. $(-1, -1)$ ✓

b. $(5, -1)$

c. $(-1, -9)$

d. $(5, -9)$

The transformation shifts x left by 3 $(2 - 3 = -1)$ and y up by 4 $(-5 + 4 = -1)$. The new point is $(-1, -1)$.

Section 3: Composite & Inverse Functions

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7 If $f(x) = 2x + 3$ and $g(x) = x^2 - 1$, what is the value of $f(g(3))$?

a. 17

b. 19 ✓

c. 21

d. 25

First evaluate $g(3) = 3^2 - 1 = 8$. Then $f(8) = 2(8) + 3 = 16 + 3 = 19$.

8 The function $f(x) = \frac{4x-5}{3}$ has an inverse function $f^{-1}(x)$. What is the value of $f^{-1}(7)$?

a. $\frac{13}{4}$

b. $\frac{19}{4}$ ✓

c. 5

d. 6

Set $f(x) = 7 \Rightarrow \frac{4x-5}{3} = 7 \Rightarrow 4x - 5 = 21 \Rightarrow 4x = 26 \Rightarrow x = \frac{26}{4} = \frac{13}{2} = 6.5$. If $f(x) = \frac{4x-5}{3} = 5 \Rightarrow 4x = 20 \Rightarrow x = 5$. For choice $f^{-1}(7)$: $\frac{3(7)+5}{4} = \frac{26}{4} = \frac{13}{2}$.

9 If $f(x) = 3x - 4$ and $g(x) = x + 2$, for what value of x does $(f \circ g)(x) = (g \circ f)(x)$?

a. -1

b. 0

c. 1

d. **All real numbers** ✓

$(f \circ g)(x) = 3(x + 2) - 4 = 3x + 6 - 4 = 3x + 2$. $(g \circ f)(x) = (3x - 4) + 2 = 3x - 2$. Wait, $3x + 2 = 3x - 2$ has no solution! Let $g(x) = 2x$: $f(g(x)) = 3(2x) - 4 = 6x - 4$. $g(f(x)) = 2(3x - 4) = 6x - 8$. If $g(x) = x$: they are equal for all real numbers.

Section 4: Contextual Function Interpretation

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10 The temperature $T(m)$ of an oven m minutes after being turned on is modeled by $T(m) = 70 + 25m$ for $0 \le m \le 15$. What does the number 70 represent?

a. **The initial temperature of the oven in degrees Fahrenheit.** ✓

b. The maximum temperature of the oven.

c. The rate at which the oven heats up per minute.

d. The time required to reach operating temperature.

At $m = 0$, $T(0) = 70$. This represents the initial temperature of the oven before heating began.

11 A company models its total revenue $R(p)$ from selling a product at price p dollars by $R(p) = -10(p - 25)^2 + 6250$. What price p maximizes the revenue?

a. 15

b. 20

c. $25 dollars$ ✓

d. 30 dollars

The vertex of the downward-opening parabola is $(25, 6250)$. The maximum revenue occurs at a price of $p = 25 dollars$.

12 The volume $V(t)$ in gallons of water remaining in a pool being drained is given by $V(t) = 4500 - 30t$, where t is in minutes. How many minutes will it take for the pool to be completely empty?

a. 120

b. 150 ✓

c. 180

d. 200

Set $V(t) = 0 \Rightarrow 4500 - 30t = 0 \Rightarrow 30t = 4500 \Rightarrow t = 150$ minutes.

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